

1.2 Sustainable Management, Design and Development

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What this lecture covers

- Sustainable management and development
- The concept of Design for Environment – DfE
- Simple, motivating examples

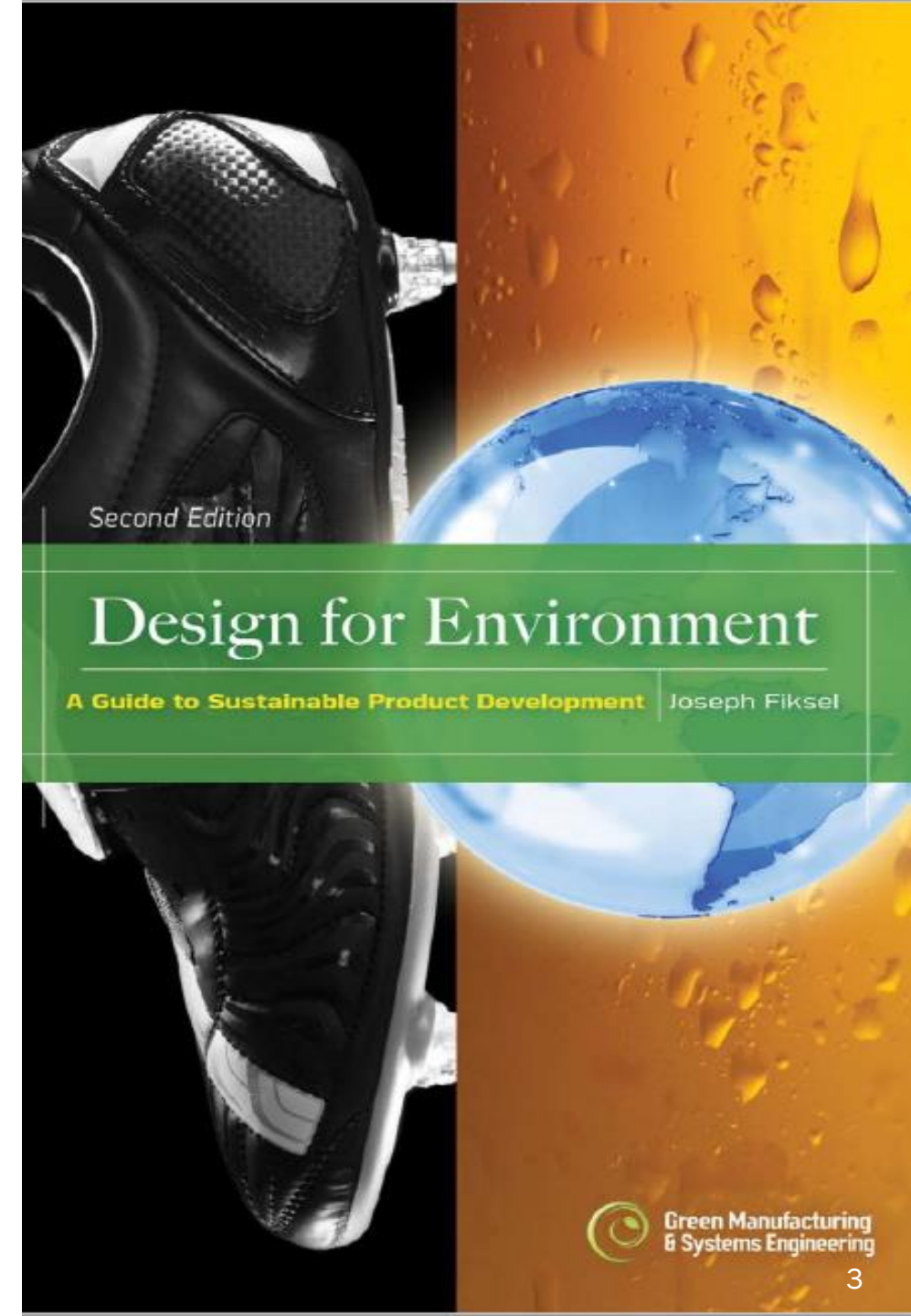


Design for Environment

A Guide to Sustainable
Product Development

By: Joseph Fiksel

Second Edition



Environmental Challenges of Today

1. Climate change, global warming and Ozone depletion
 - *Hotter temperatures, more extreme weather events*
2. Air pollution
 - *Haze, acid rain*
3. Desertification
 - *Overgrazing, deforestation*
4. Pollution due to waste disposal
 - *Polluted water due to leached pesticides and herbicides*
5. Loss of biodiversity
 - *Extinction of species*



How Do We Find a Solution?

What is Sustainable Management*?

- Management that meets the needs of the present without compromising the ability of future generations to meet their own reasonably foreseeable needs
- This includes at least the future generations within the anticipated lifetime of the end products and by-products of those activities.

*Engineers New Zealand definition based on “Report of the World Commission on Environment and Development: Our Common Future, 1987”

A Return to Ancient Values

- Western culture often views the universe, including the environment, as a complex, logical and ordered system; one to “conquer”
- Māori traditions and principles (e.g., kaitiakitanga), like other ancient civilisations, view human’s relationship with “nature” differently
- Chief Seattle, American Indian, said:

“This we know: the earth does not belong to man, man belongs to the earth. Man did not weave the web of life, he is merely a strand in it.”



Why Design for Environment (DFE)?

- Government and communities insists that decisions and projects are sustainable
- Engineers must embrace and implement sustainability through engineering practice to remain relevant
- This is a paradigm shift in the way in which engineering is practiced
- It requires an awareness and understanding of the importance of the topic, together with an analysis toolkit to support it.

External drivers

- Climate change awareness
- Governmental initiatives
- Rating & liability schemes
- Codes & standards
- Environmental advocacy

Business drivers

- Company values
- Reputation & branding
- Shareholder value creation
- Competitive pressures
- Market opportunities

Corporate
Environmental
Responsibility



What is DFE?

- Much more than a few extra design calculations!
- Need to consider different lens with which to view the problem (e.g. PESTLE Analysis)
- Engineers need to apply a **holistic design approach**

Engineers play have a critical role in creating sustainable goods and services, through to communities, cities and countries

Encouraging Sustainable Practices

1. National policy and law, e.g. *Resource Management Act*
2. International trade and relations, e.g. *free and fair trade agreements*
3. Process operations and product design decisions, e.g. *renewable energy*
4. Consumer behaviour, e.g. *recycling waste*

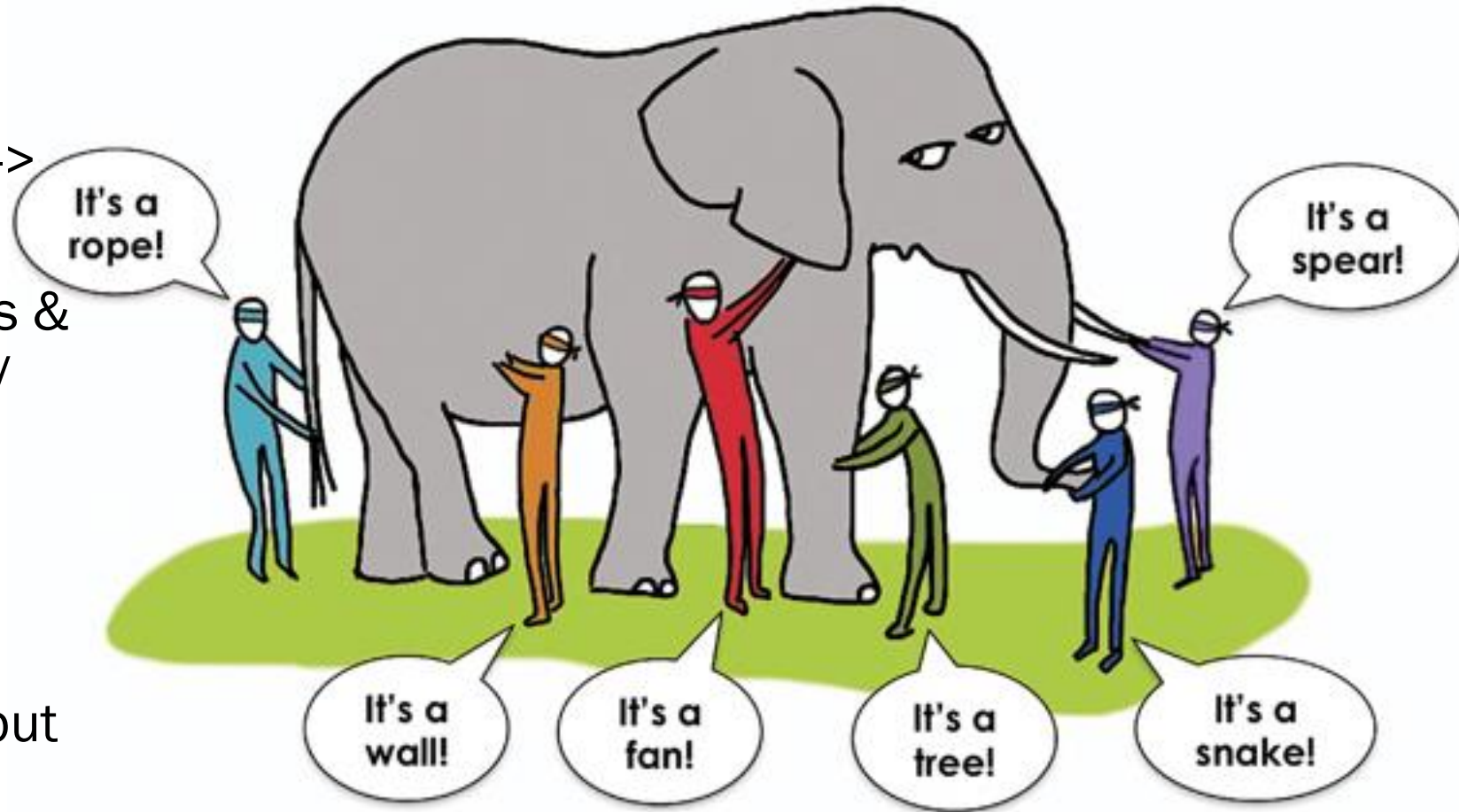
Resource Management Act

Principles and purposes:

- a) Sustain the potential of natural and physical resources (excluding minerals) to meet the reasonably foreseeable needs of future generations
- b) Safeguard the life-supporting capacity of air, water, soil, and ecosystems
- c) Avoid, remedy, or mitigate any adverse effects of activities on the environment

The Challenge of Sustainability

- Complex interactions -> difficult to quantify
- Numerous interactions & impacts -> uncertainty
- Sustainable design/solutions are non-obvious
- Impacts might be obvious in hindsight, but hard to forecast



Solving design challenges requires the full picture of sustainability

Preventative vs Reactive Design Solutions

Preventative measures, e.g.

- a. Replacing fossil fuel use with renewable energy
- b. Developing sustainable materials, e.g. “green” concrete
- c. Recycling waste material , e.g. plastics, paper, metals
- d. Using scarce water resources more efficiently

Reactive measures, e.g.

- a. Building seawalls to defend against sea level rise
- b. Developing drought-tolerant crops
- c. Adapting building codes to consider extreme weather events

Design for Environment – Not So Easy

- Hindsight is a wonderful thing, but as engineers, we need to have the vision to avoid such failures
- Everything is a trade-off in engineering
- Let's look at three examples...
 - *Paper vs polystyrene cups*
 - *Plastic, paper and reusable bags*
 - *Bitcoin*

Which is more environmentally-friendly?



Paper



Polystyrene

Are paper cups more environmentally-friendly?

Footprint	Paper cup	Polystyrene cup
Primary feedstock per cup	33 g of wood	3.2 g of oil
Required oil per cup	4.1 g of oil (transportation)	3.2 g of oil
Chemical waste per cup	1.8 g of Chlorine, sodium hydroxide, bleach, sulphuric acid, sulphur and limestone	0.05 g
Relative water use	580	1
Reusable	No	Yes, a few times
Energy value at end of life	20 MJ/kg	40 MJ/kg



Are plastic bags bad for the environment?

Basis	1,500 kg	1,000 kg	1,000 kg	1,000 kg
				
	Polyethylene (mass = 6 kg)	Paper (52 kg)	Reusable nonwoven polypropylene (42 kg)	Reusable polyethylene with 40% recycled content (44 kg)
	1 use	1 use	1 use	8 uses
Nonrenewable energy, GJ	763	2,620	3,736	467
Greenhouse gas emissions metric tons of CO ₂ equivalent ● = 0.010	0.040	0.080	0.262	0.033
Freshwater consumption, gal 💧 = 10 (1 gal = 4 L)	58	1,000	426	85
				250
				40



Is Bitcoin bad for the environment?

In 2018, the Bitcoin network consumed 31.29 TWh of electricity and had a carbon footprint of 17.3 Mt CO₂-eq

(NZ emitted 55.5 CO₂-eq)

Environment vs financial security:
how does one decide?

Susanne Köhler and Massimo Pizzol, 2019. Life Cycle Assessment of Bitcoin Mining. *Environmental Science & Technology* 53 (23), 13598-13606.

Summary

- There are a variety of environmental challenges, each of which are complex and multi-faceted
- Sustainable solutions and design must reach beyond merely engineering alone, e.g., PESTLE analysis
- Choosing the most environmentally friendly solution is non-trivial, indicators are often in tension with trade-offs
- This course teaches a set of design guidelines and analysis methods to help you think critically about engineering and the environment